CS224
Lab No. 7
Section No. 4
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(a )
Special Function Registers (SFRs) for DC Motor (Part 2-b)
- TRISE: Configures the direction of PORTE pins (input for push buttons RE0 and RE1)
- TRISA: Configures the direction of PORTA pins (output to control the motor direction using RA0 and RA1)
- PORTE: Used to read button states (RE0 for clockwise, RE1 for counterclockwise)
- LATA: Used to write logic levels to output pins (RA0 and RA1) to control motor direction
(b )

```
void Wait() {
 Delay_ms(1000);
}


void main() {
 AD1PCFG = 0xFFFF;
 DDPCON.JTAGEN = 0;
 TRISE = 0XFFFF;
 TRISA = 0X0000;




 LATA.F0 = 0;
 LATA.F1 = 0;
 LATA.F2 = 0;


 Wait();
```

```
 while(1)
 {
        unsigned int btn0 = PORTE.F0;
        unsigned int btn1 = PORTE.F1;


        if(btn0 ^ btn1){
            Wait();


            if(btn1 == 1){
              LATA.F0 = 1;
              LATA.F1 = 1;
              LATA.F2 = 0;
            }
            else{
                LATA.F0 = 0;
                LATA.F1 = 0;
                LATA.F2 = 1;
            }


            Wait();
            LATA.F0 = 0;
            LATA.F1 = 0;
            LATA.F2 = 0;
        }


        else
        {
           LATA.F0 = 0;
           LATA.F1 = 0;
           LATA.F2 = 0;
        }
}
}
```

(c )

- TRISA: Configures PORTA as output (used to drive segment lines a–g)
- TRISE: Configures PORTE as output (used to enable digits)

- PORTA: Used to output the binary pattern for each digit segment
- PORTE: Used to select which digit to activate (only one active at a time

(d )

```
/*
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Seven-Segment Display: Fibonacci Sequence Visualization

Displays each Fibonacci number with a wait proportional to its value.
Each number is broken into 4 digits and shown using multiplexing.
*/

unsigned char segmentMap[10] = {
    0x3F, // 0
    0x06, // 1
    0x5B, // 2
    0x4F, // 3
    0x66, // 4
    0x6D, // 5
    0x7D, // 6
    0x07, // 7
    0x7F, // 8
    0x6F  // 9
};

unsigned int prevVal = 0;
unsigned int currVal = 1;
unsigned int displayCounter = 0;

unsigned int d1 = 0; // leftmost digit
unsigned int d2 = 0;
unsigned int d3 = 0;
unsigned int d4 = 0; // rightmost digit

void setup() {
    AD1PCFG = 0xFFFF;    // All analog pins to digital
    JTAGEN_bit = 0;      // Release pins from JTAG
```

```
    TRISA = 0x00;       // PORTA as output (segments A-G)
    TRISE = 0x00;       // PORTE as output (digit selectors)
}


void displayNumber(unsigned int num, unsigned int duration_ms) {
    // Break the number into 4 digits
    d1 = num / 1000;
    d2 = (num / 100) % 10;
    d3 = (num / 10) % 10;
    d4 = num % 10;


    // Multiplex the display for num * 100 ms total
    while (displayCounter < num * 100) {
        // Digit 1
        PORTA = segmentMap[d1];
        PORTE = 0x01;
        Delay_ms(1);


        // Digit 2
        PORTA = segmentMap[d2];
        PORTE = 0x02;
        Delay_ms(1);


        // Digit 3
        PORTA = segmentMap[d3];
        PORTE = 0x04;
        Delay_ms(1);


        // Digit 4
        PORTA = segmentMap[d4];
        PORTE = 0x08;
        Delay_ms(1);


        displayCounter++;
    }
```

```
    displayCounter = 0;
}


void main() {
    setup();


    while (1) {
        displayNumber(currVal, currVal * 100);


        // Fibonacci update
        currVal = currVal + prevVal;
        prevVal = currVal - prevVal;
    }
}
```